HomeRelay

Technical Overview & Architecture

From household situations to coordinated next steps
An MCP-powered household coordination agent that connects situations, tasks, documents, and proposed plans while keeping the user in control.

245	36/36	28	0
tests reported passing	assistant evaluations	browser tests	audit vulnerabilities reported

Project at a glance

- **Purpose:** reduce fragmented household coordination by turning everyday events into structured plans and trackable tasks.
- **Architecture:** React + Vite frontend, Express API, dedicated MCP server, TypeScript domain logic, and SQLite persistence.
- **Agent modes:** deterministic rules-based mode plus an optional Anthropic-powered mode. Core functionality can run without an external LLM API key.
- **Current scope:** a local, single-user prototype with a remote MCP/OAuth profile and containerized deployment workflow.

Product capabilities

HomeRelay keeps household context in one persistent workspace, allowing the agent to connect a situation to relevant tasks, documents, and proposed actions.

Capability	What it provides
Situation-aware planning	Capture deliveries, maintenance visits, family events, and other situations; produce a structured plan for review.
Human-approved actions	Keep a review-and-approval step between proposed plans and operational task creation.
Adaptive replanning	Revise plans as circumstances change and prevent approval of plans tied to resolved situations.
Task management	Organize due dates, priorities, dependencies, and recurring work.
Document vault	Upload, search, and associate household documents with relevant tasks and situations.

Capability	What it provides
Relay Map and activity	Explore relationships among household records and review recorded decisions and changes.

Architecture and design choices

- **Frontend:** React and Vite provide the user-facing workspace.
- **API layer:** Express coordinates application requests.
- **Tool boundary:** a dedicated MCP server exposes structured capabilities; the agent uses tools rather than directly manipulating the database.
- **Persistence:** SQLite stores household state across application restarts.
- **Agent flexibility:** deterministic rules mode supports basic operation without requiring an external model API key; optional Anthropic mode can provide LLM-based behavior when configured.

Reliability and security

- **Concurrent uploads:** duplicate-upload handling and transactional checks strengthen consistency during simultaneous uploads.
- **Approval consistency:** plans associated with resolved situations cannot be approved into active work.
- **Storage controls:** configurable document-storage quota.
- **Web security:** Content Security Policy and security headers for the production preview.
- **Benchmark validity:** failed and rate-limited requests are excluded from successful performance measurements.
- **Remote access:** OAuth-protected remote MCP access with permission boundaries for sensitive and destructive actions.

Testing and engineering

The following results were reported by the project developer for the current implementation.

Area	Reported outcome
Type checking and production build	Passing
Unit and integration tests	245 passing
Assistant evaluations	36/36 passing
Browser end-to-end tests	28 passing
Dependency audit	0 vulnerabilities reported
Docker over HTTP	47 checks exercised
Docker over HTTPS	42 checks exercised, including authentication and token behavior

Security and control principles

- **Human oversight:** proposed plans are reviewed before they create operational tasks.
- **Least privilege:** remote tool permissions restrict sensitive actions.
- **State consistency:** transactional validation protects important changes and approval decisions.
- **Operational visibility:** activity history records changes and distinguishes simulated events from real-world confirmations.

Roadmap

- **Alexa+:** complete the integration through Amazon's official developer workflow once the required preview/CLI access is approved.
- **Public endpoint:** configure a production domain and hosting, then verify remote access over public HTTPS.
- **Policy readiness:** finalize operator details and obtain appropriate review of privacy and terms.
- **Reliability:** continue extended reliability testing as part of release readiness.

Summary
HomeRelay combines persistent household context, structured MCP tools, situation-aware planning, human approval, document/task relationships, and automated testing into a modular household coordination prototype.

Prepared from project information supplied for the hackathon submission.